INDIAN INSTITUTE OF TECHNOLOGY PATNA

EC3202: Digital Signal Processing Lab

EXPERIMENT NO: 03

Z Transform

Submitted by -

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Aim 1:

To understand the Z-transform operations and solve the differential equations.

1. Let X1(z) = 2 + 3z−1 + 4z−2 and X2(z) = 3 + 4z−1 + 5z−2 + 6z−3. Determine X3(z) = X1(z)X2(z).
2. Determine the inverse z-transform of

$$X(z) = \frac{1 + 0.4\sqrt{2}z^{-1}}{1 - 0.8\sqrt{2}z^{-1} + 0.64z^{-2}}$$

   so that the resulting sequence is causal and contains no complex numbers.
3. Given a causal system y(n) = 0.9y(n − 1) + x(n)
   a. Determine H(z) and sketch its pole-zero plot.
   b. Plot |H(ejω)| and H(ejω).
   c. Determine the impulse response h(n).
4.

Solve the difference equation

$$y(n) = \frac{1}{3}\left[x(n) + x(n-1) + x(n-2)\right] + 0.95y(n-1) - 0.9025y(n-2), \quad n \geq 0$$

where $x(n) = \cos(\pi n/3)u(n)$ and

$$y(-1) = -2,\ y(-2) = -3; \quad x(-1) = 1,\ x(-2) = 1$$

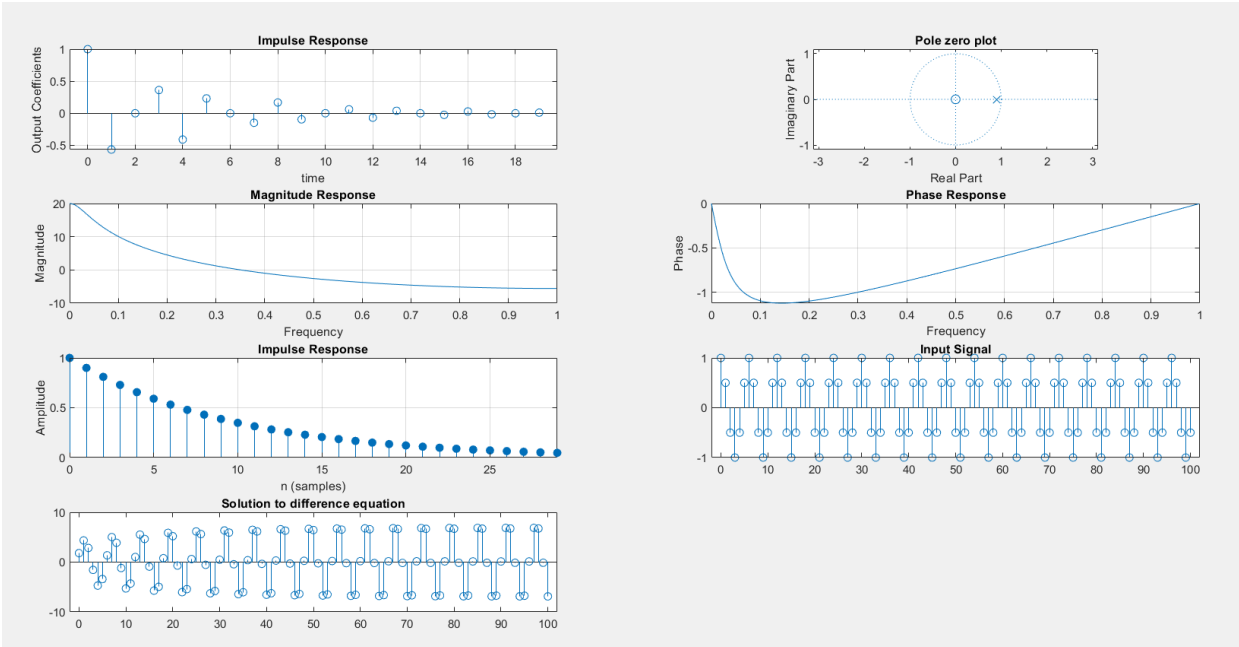
First determine the solution analytically and then by using MATLAB.

MATLAB Code:

```
%Q - 1
X1 = [2 3 4];
X2 = [3 4 5 6];
X3 = conv(X1,X2);
disp(X3);

%Q - 2
syms z n
X = (1+(0.4*sqrt(2)*z^(-1)))/(1-(0.8*sqrt(2)*z^(-1))+(0.64*z^(-2)));
H = iztrans(X,z,n);
pretty(H);  disp(H);

X_num = [1 0.4*sqrt(2)];
X_den = [1 0.8*sqrt(2) 0.64];
[H1,t] = impz(X_num,X_den,20);
subplot(4,2,1);  stem(t,H1);
xlabel("time"); ylabel("Output Coefficients");
```

```
title("Impulse Response"); grid on;

%Q - 3
Y_coeff = [1 -0.9];
X_coeff = 1;
subplot(4,2,2);
zplane(X_coeff,Y_coeff);
title("Pole zero plot");
[H2,W] = freqz(X_coeff,Y_coeff,512); subplot(4,2,3);
plot(W/pi,20*log10(abs(H2)));
title("Magnitude Response");
xlabel("Frequency");
ylabel("Magnitude");  grid on;

subplot(4,2,4);
plot(W/pi,angle(H2));
title("Phase Response");
xlabel("Frequency"); ylabel("Phase");
grid on;

subplot(4,2,5);
impz(X_coeff,Y_coeff,30);
title("Impulse Response"); grid on;

%Q - 4
n = 0 : 100 ;
X_coeff1 = [1/3 1/3 1/3];
Y_coeff1 = [1 -0.95 0.9025];
X_n = cos(pi*n/3);
Y_initial = [-2 -3];
X_initial = [1 1];
Z_i = filtic(X_coeff1,Y_coeff1,Y_initial,X_initial);
Y = filter(X_coeff1,Y_coeff1,X_n,Z_i);
subplot(4,2,6);
stem(n,X_n); title("Input Signal");
subplot(4,2,7); stem(n,Y);
title("Solution to difference equation");
```

Plot:



Command Window

```
>> untitled
     6    17    34    43    38    24

   n            1 - n                        n - 1
(-1)  sqrt(2) 25      (sqrt(2) (- 10 - 10i))      4i
-----------------------------------------------------
                         5

                                                   n   n    / 3 pi n \
        n           1 - n                       n - 1      (-1)  16  cos| ------ | 3
    (-1)  sqrt(2) 25      (sqrt(2) (- 10 + 10i))      4i                \    4   /
  - ----------------------------------------------------- + ---------------------------
                                 5                                      n
                                                                      20

((-1)^n*2^(1/2)*25^(1 - n)*(2^(1/2)*(- 10 - 10i))^(n - 1)*4i)/5 - ((-1)^n*2^(1/2)*25^(1 - n)*(2^(1/2)*(- 10 + 10i))^(n - 1)*4i)/5 + (3*(-1)^n*16^n*cos((3*pi*n)/4))/20^n
```

Result:

We have successfully done each part of the experiment and matched the data with calculated values and understood the operation.